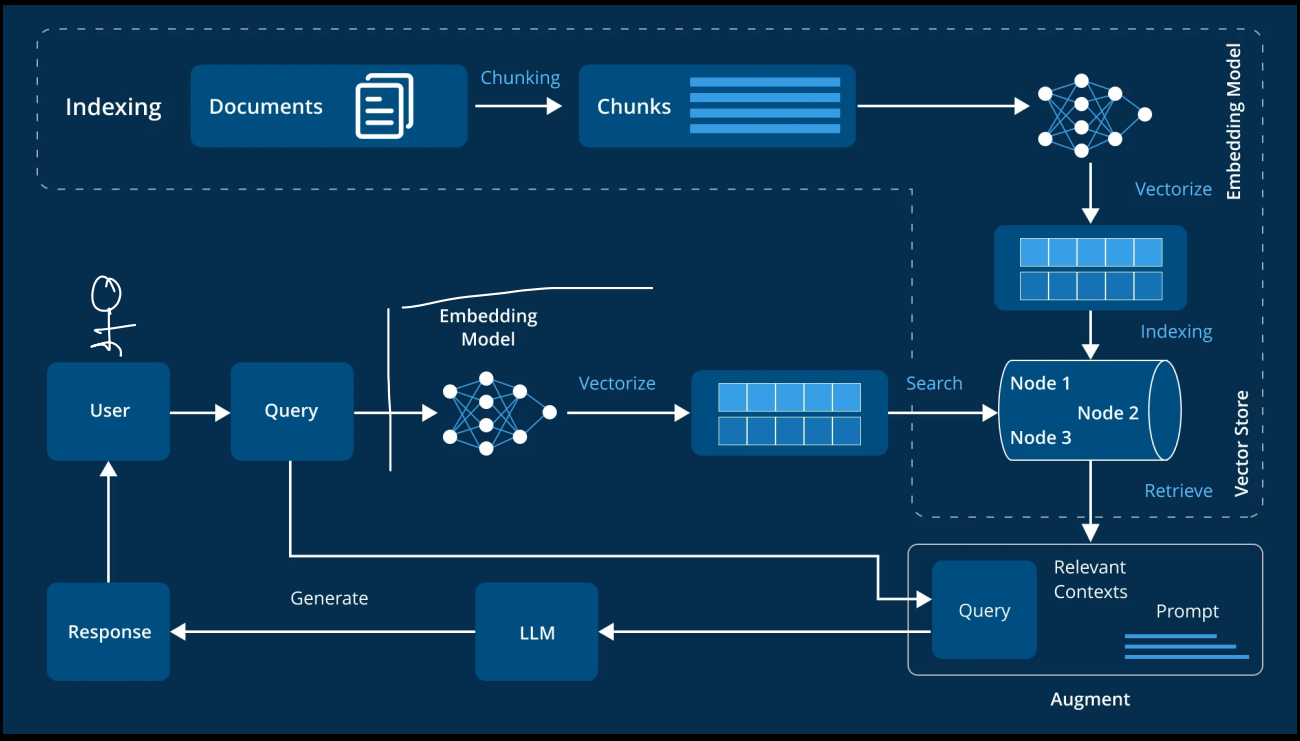
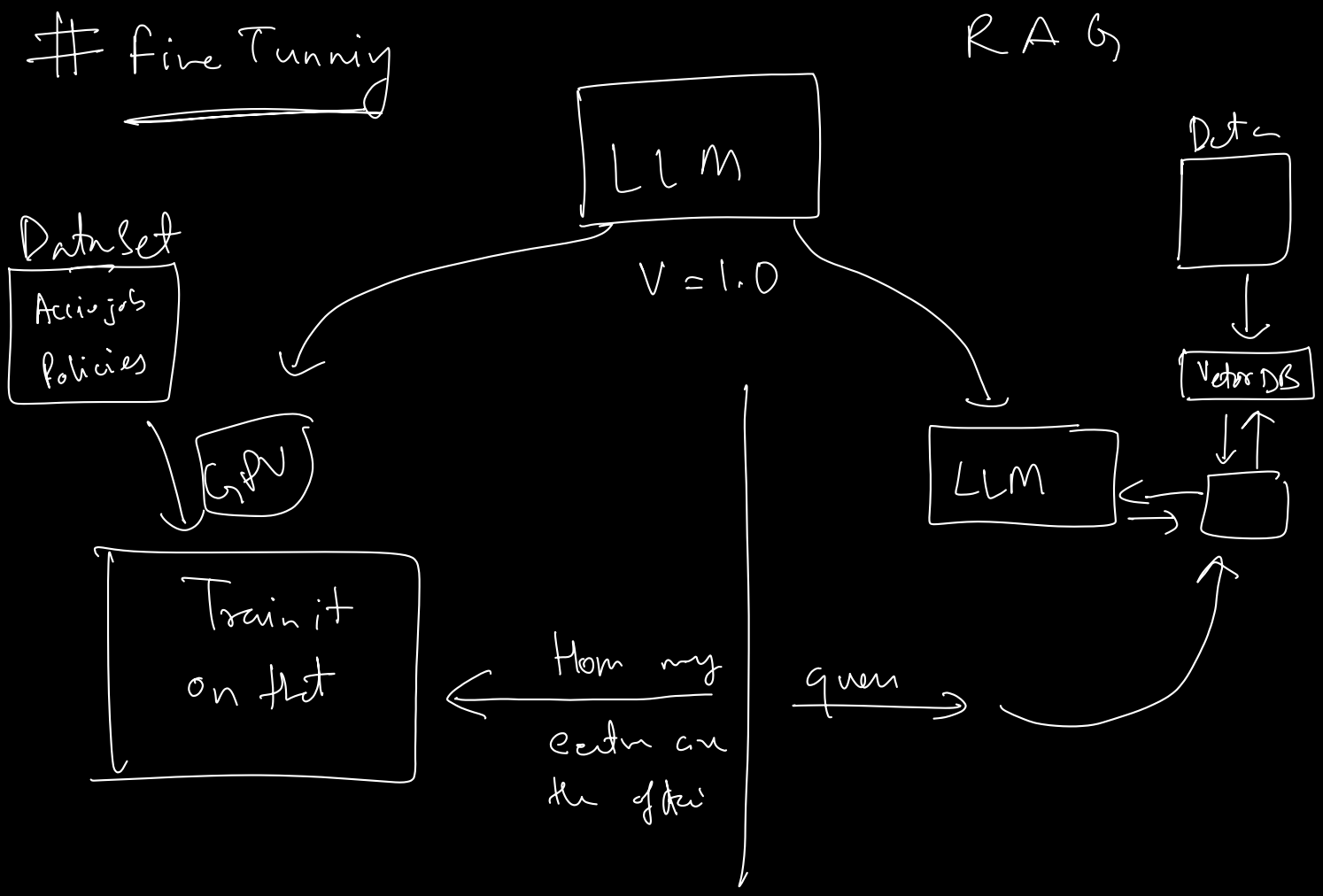
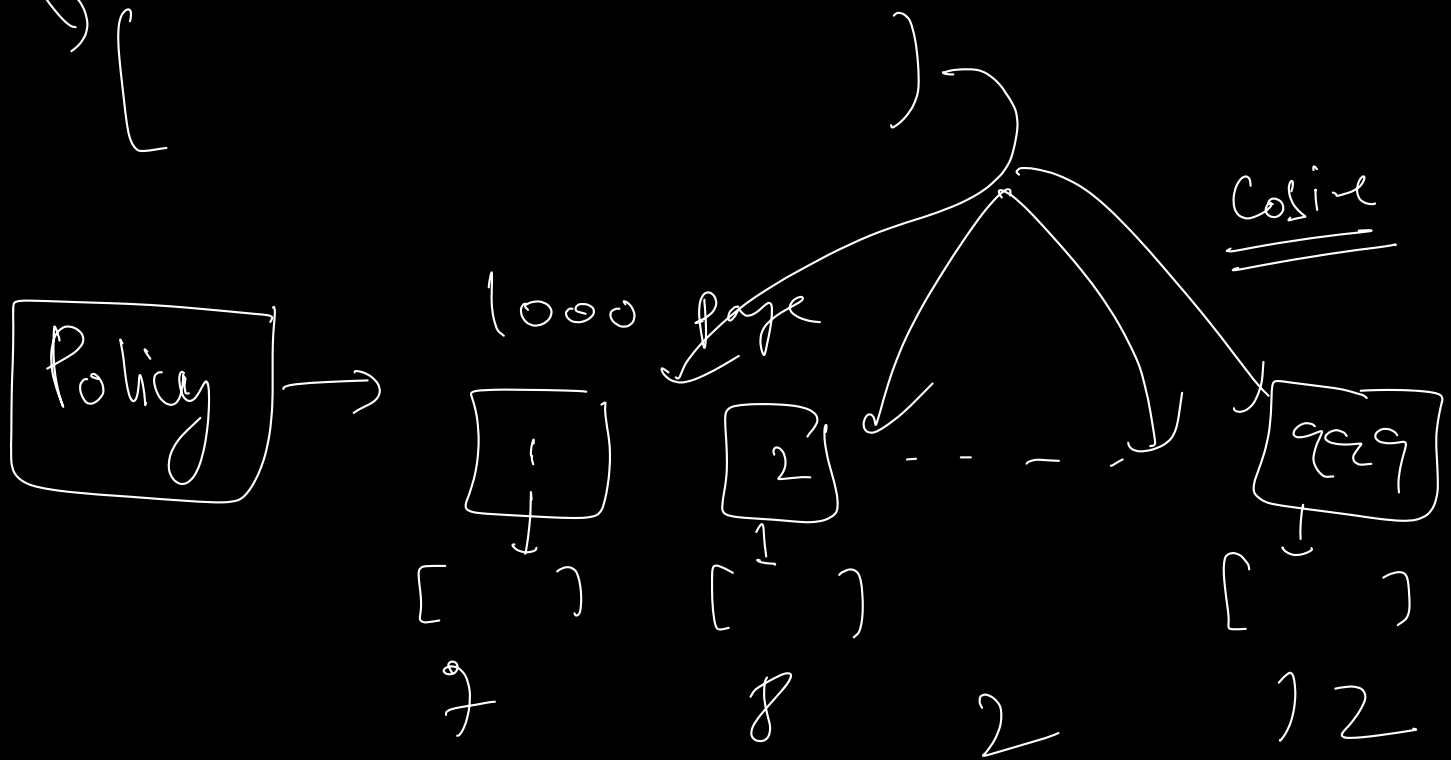

Indexing
Documents
Chunking
Chunks
Embedding Model
Vectorize
Indexing
User
Query
Embedding Model
Vectorize
Search
Node 1
Node 2
Node 3
Vector Store
Retrieve
Query
Relevant Contexts
Prompt
Augment
Response
Generate
LLM


Fine Tunning
RAG
LLM
V=1.0
Data
DataSet
Activities
Policies
GPU
Vector DB
LLM
Train it
on that
How my
system can
the data
query

[How many offline centres.] vector



top-5

12 [ ]
11 [ ]
10 [ ]

id —

Vector [ ]

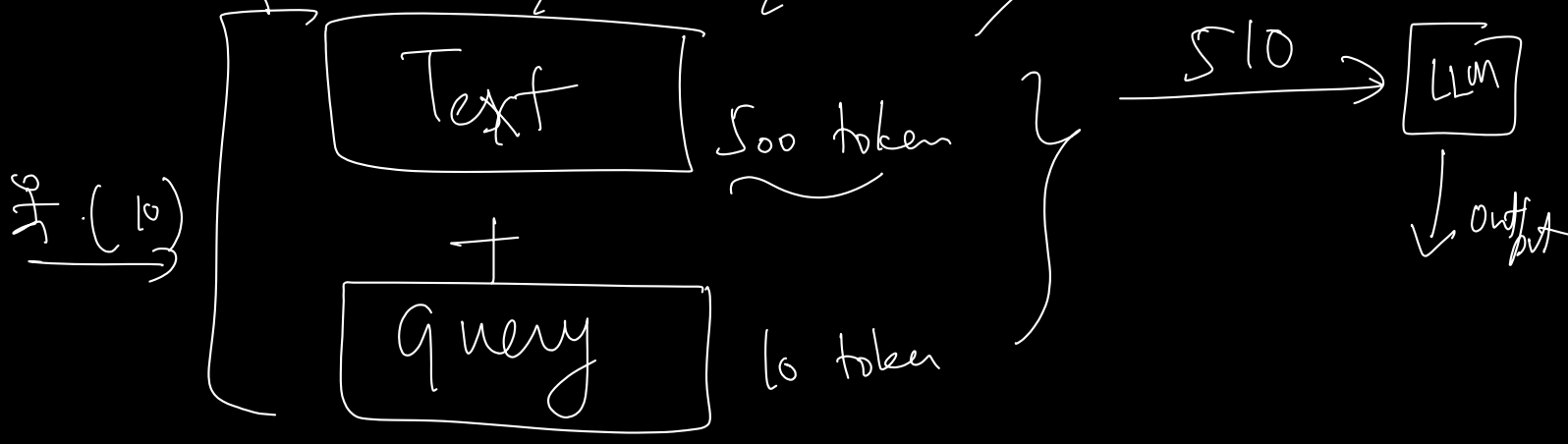
meta data — [text]

5 pages

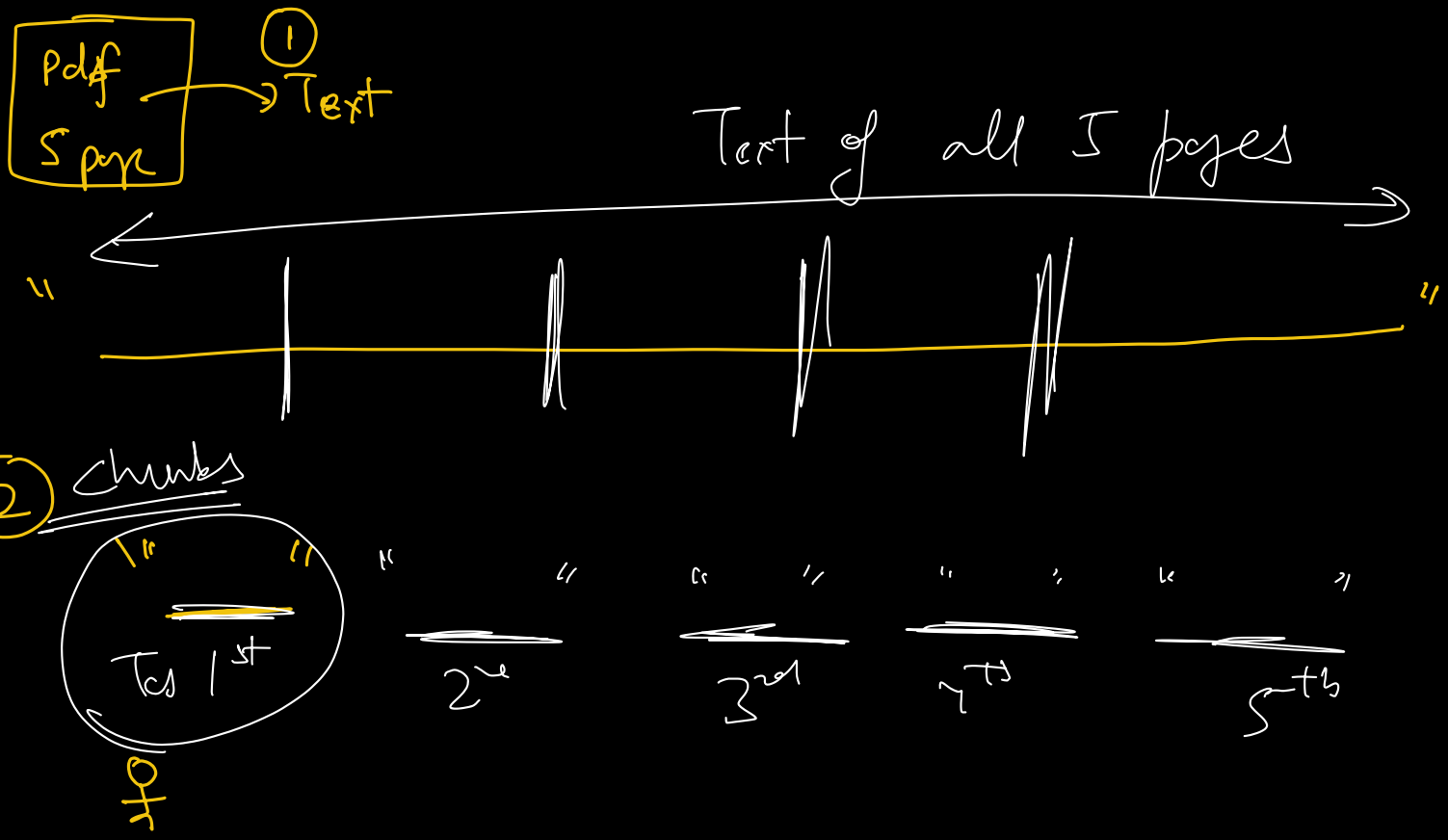
Thi ≃ 1000

1 user

Pg 13 | Pg 14 | Pg 50 | Pg 80 | Pg 85



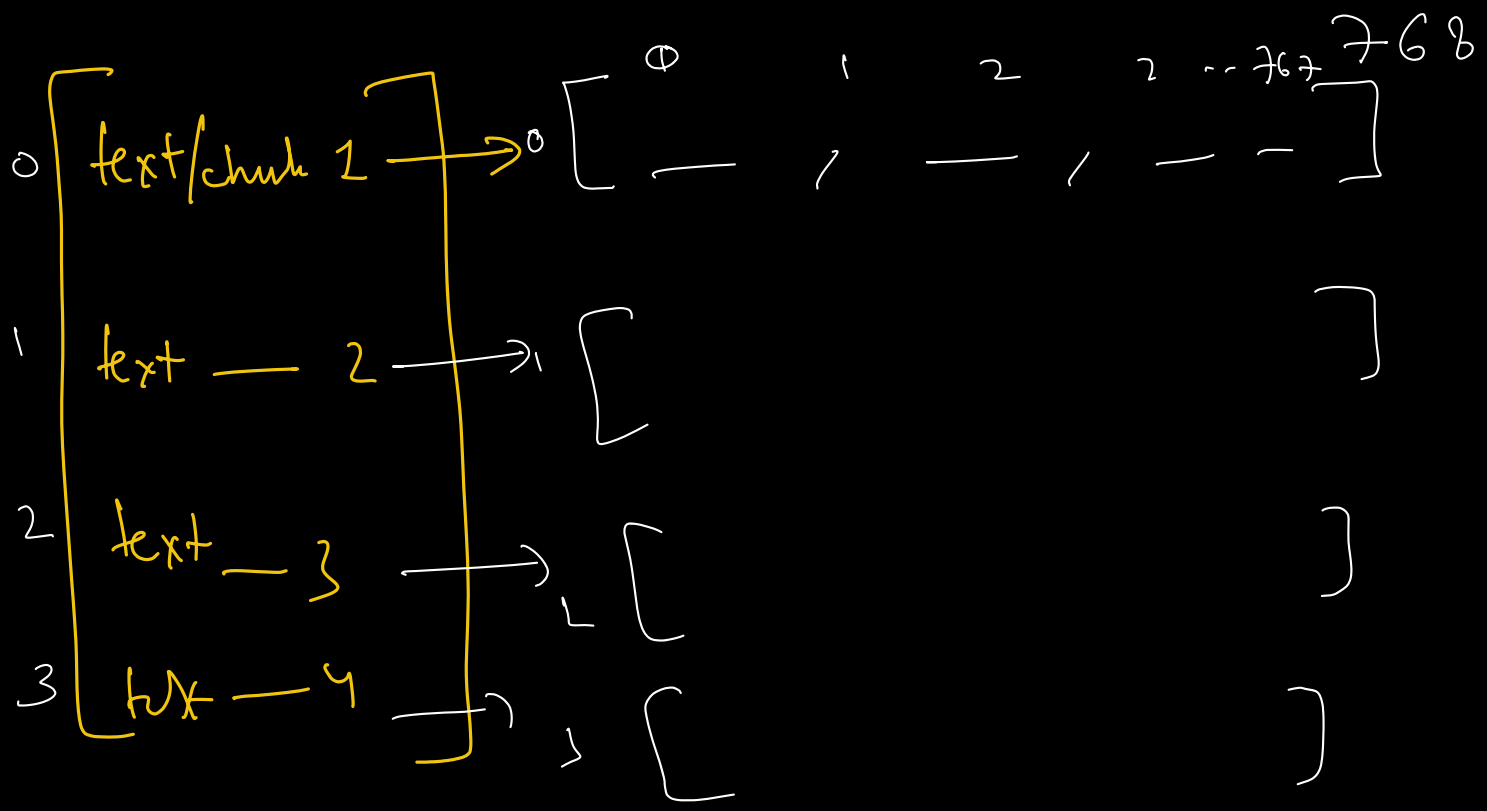
768



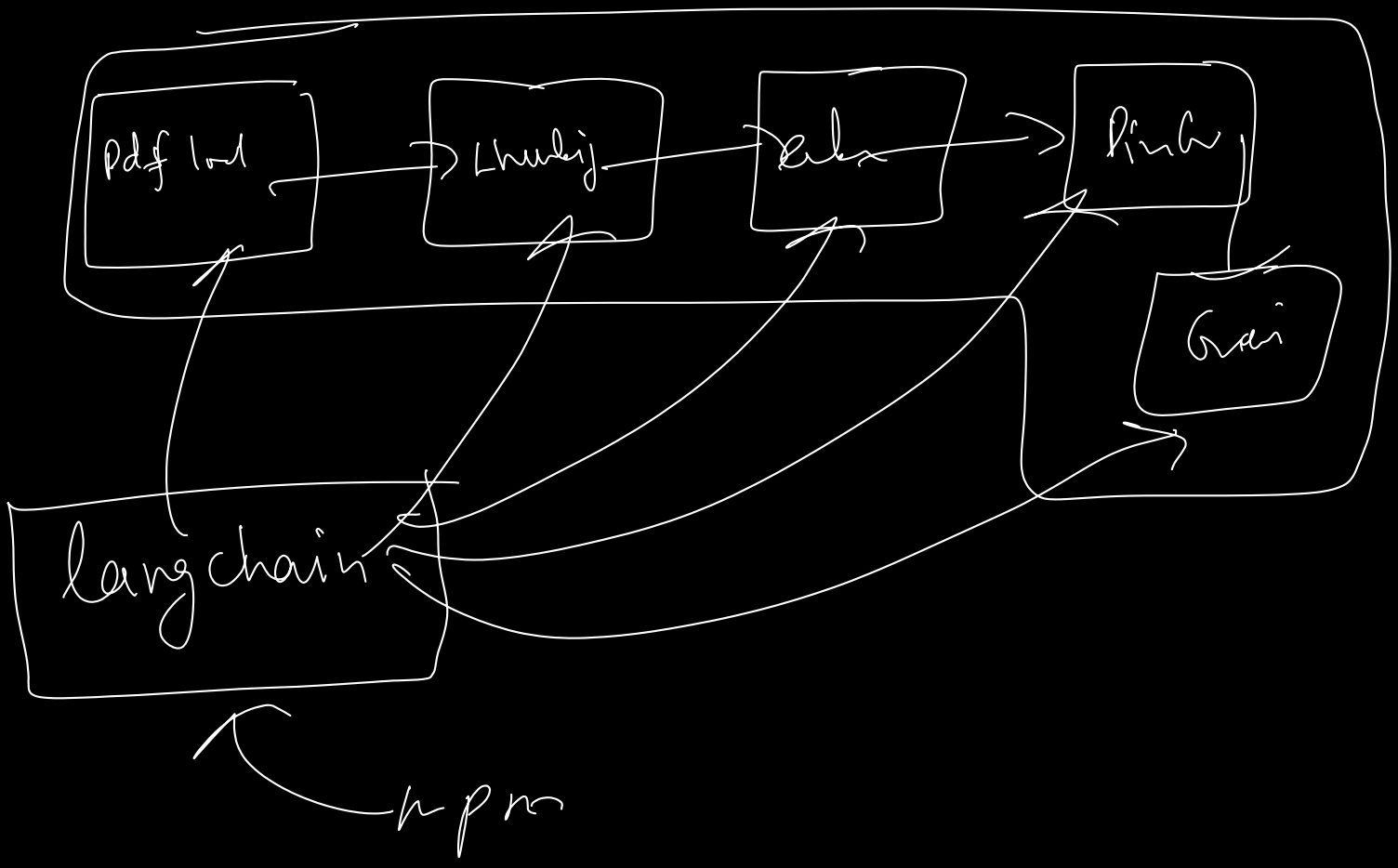
MERN

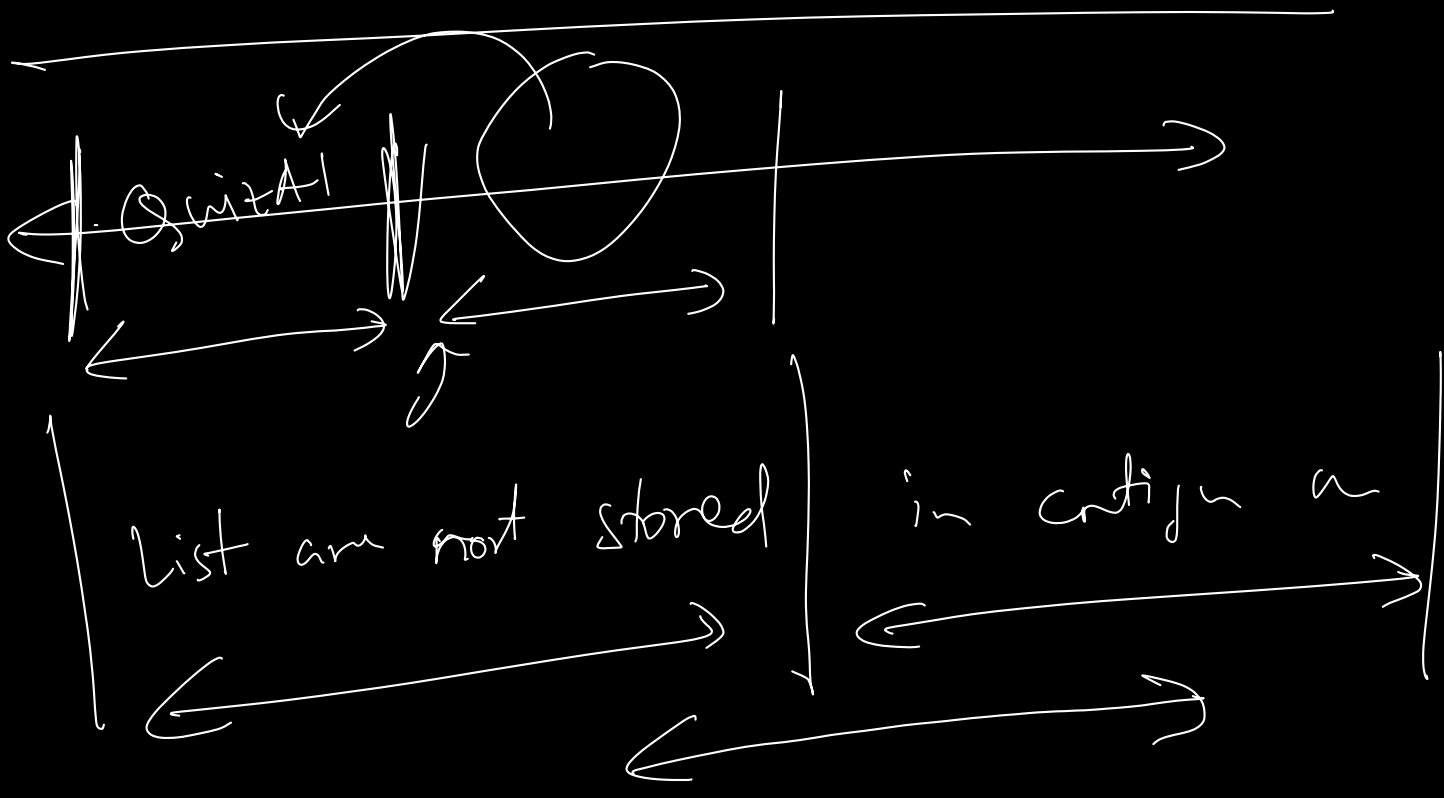
3 Vector for Each Chunk

4 Store them in Vector DB

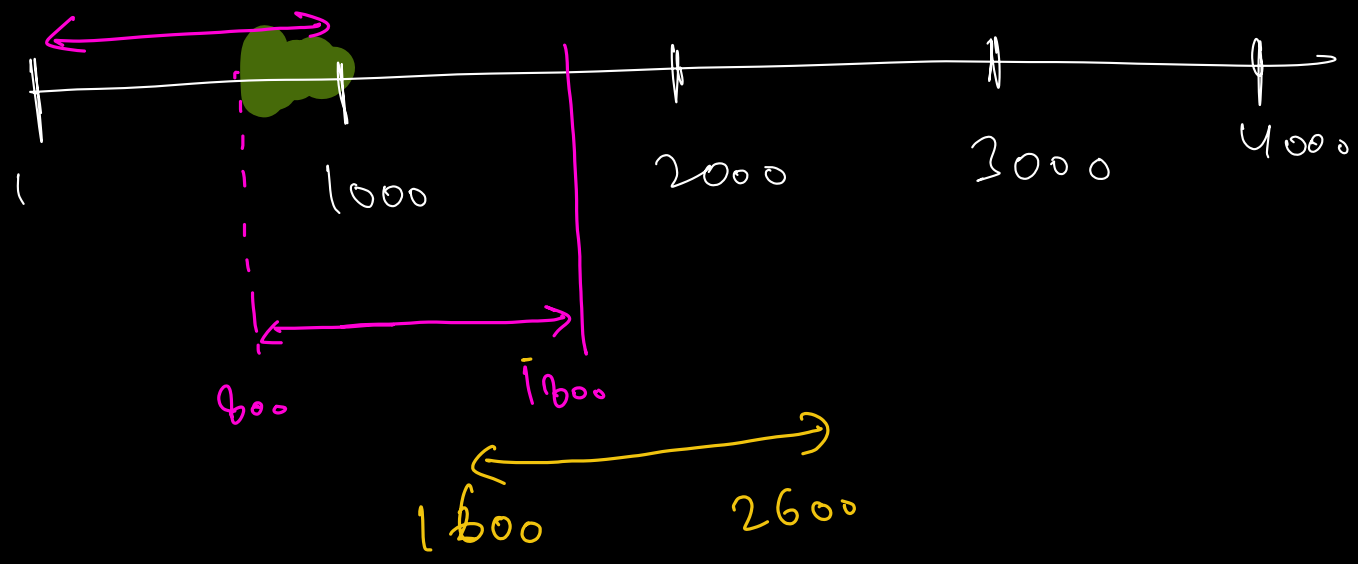


ch


Pdf load
Chunking
Embed
Pinecone
Groq
langchain
npm

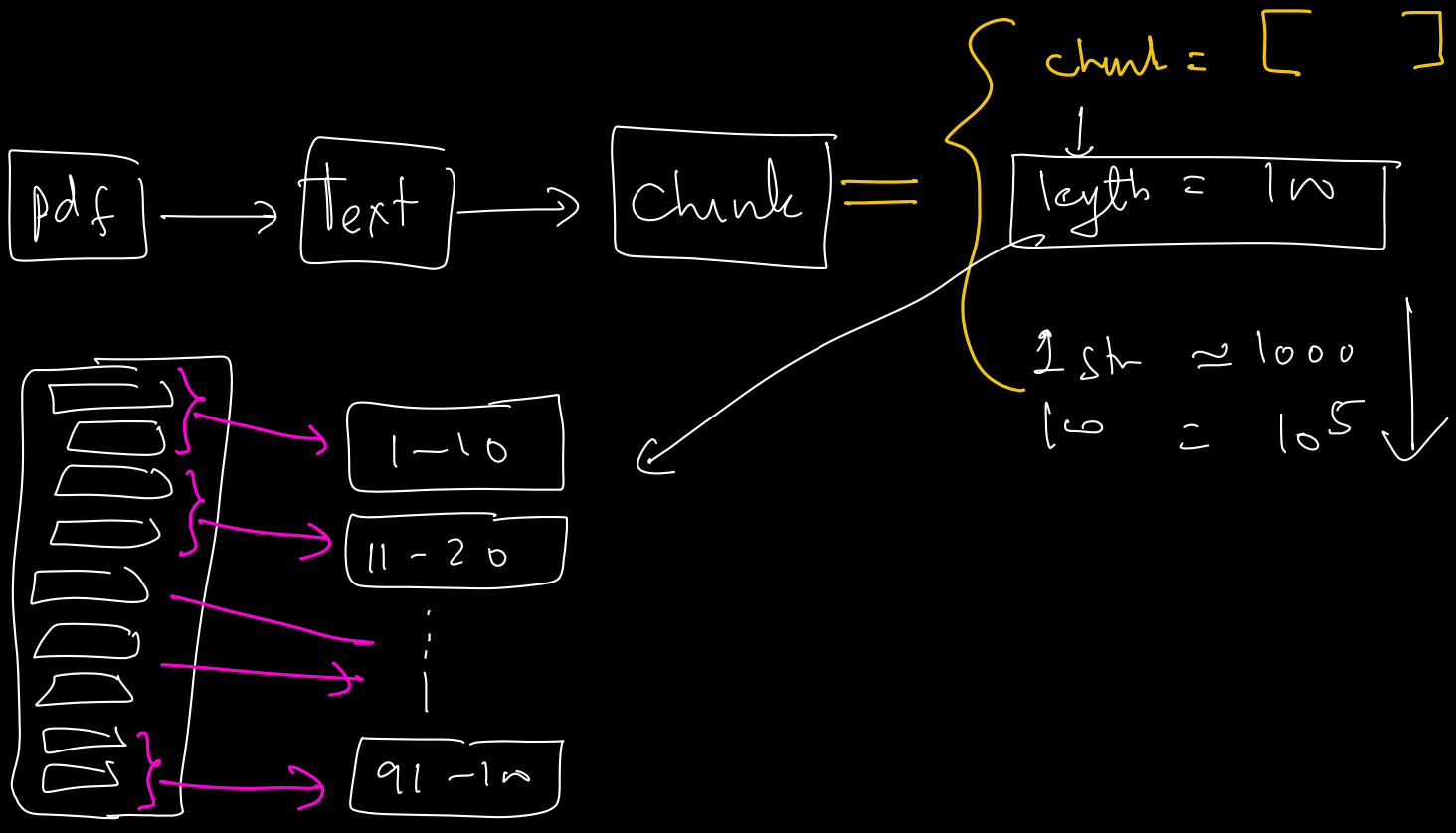

OpenAI
List are not stored in antigen a

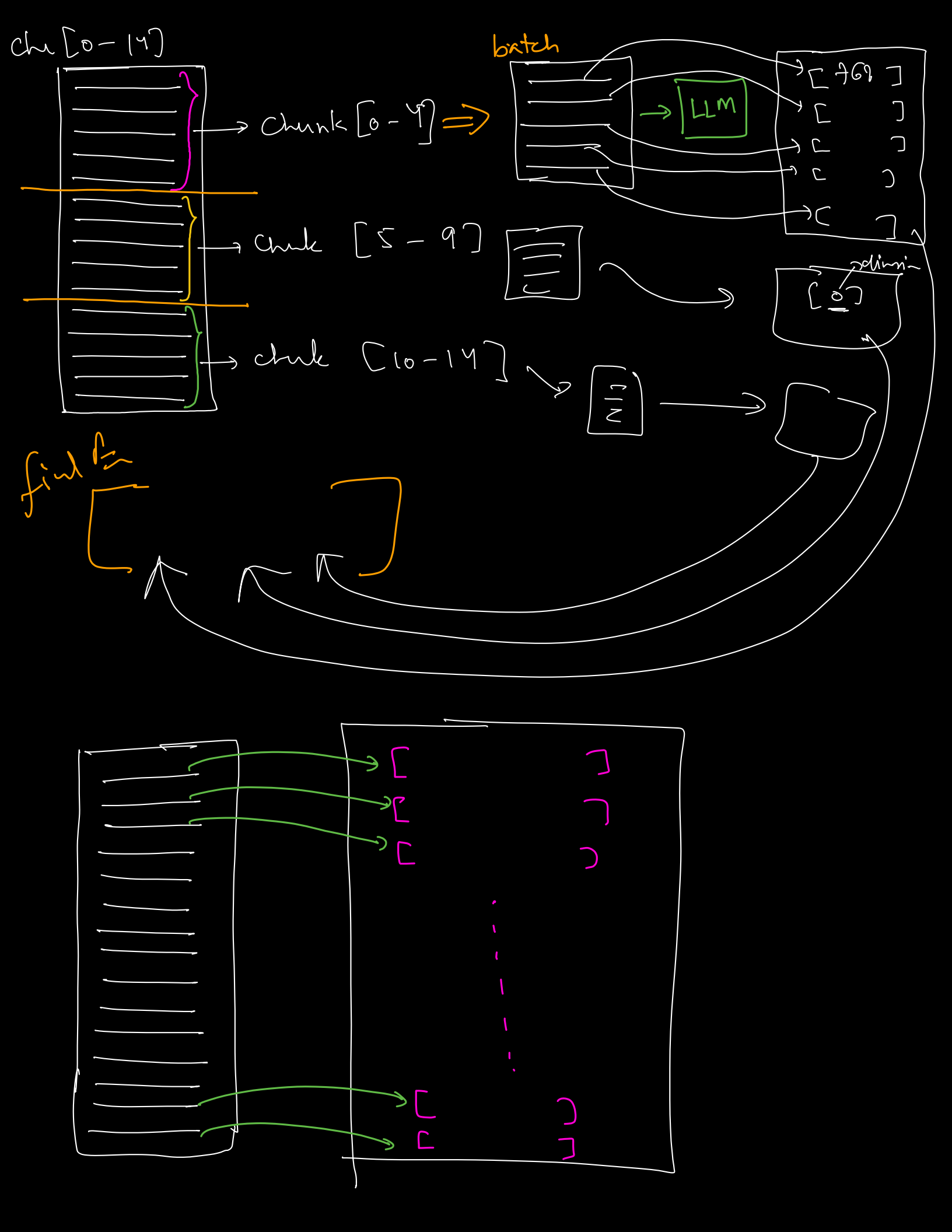
100

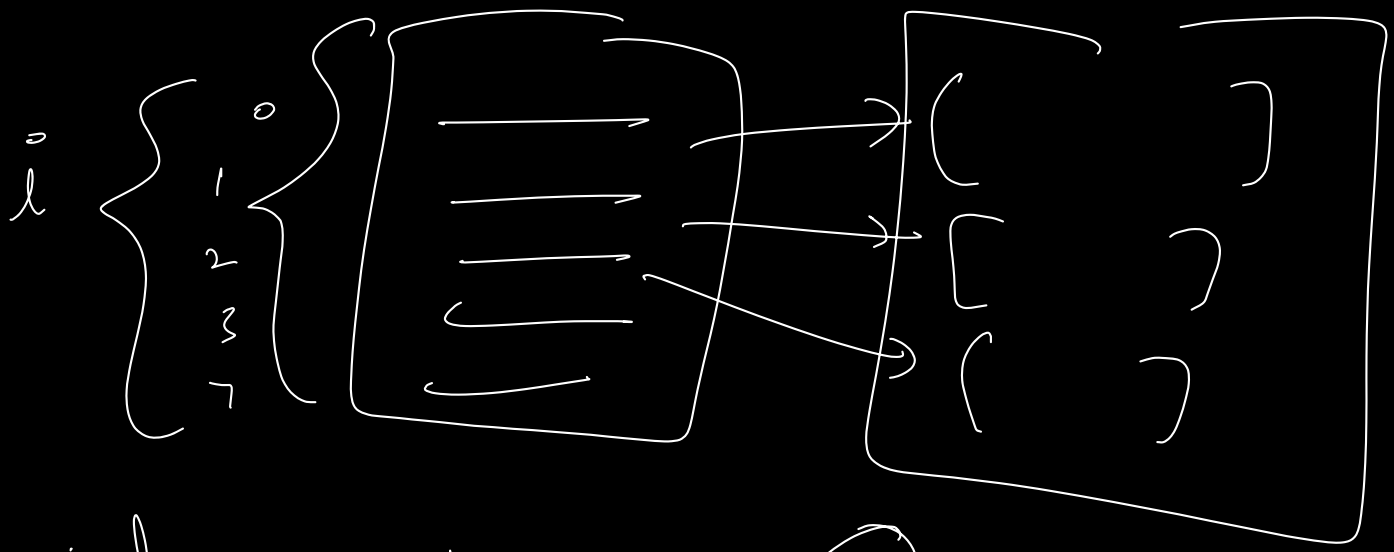

1
1000
2000
3000
4000
800
1800
1600
2600

$$f(t, x, y)$$

fixed


pdf
text
Chunk
chunk = [ ]
length = 1m
1st ≃ 1000
1e5 = 10^5
1-10
11-20
91-1m


chu [0-14]
chunk [0-4]
batch
LLM
chunk [5-9]
chunk [10-14]
find

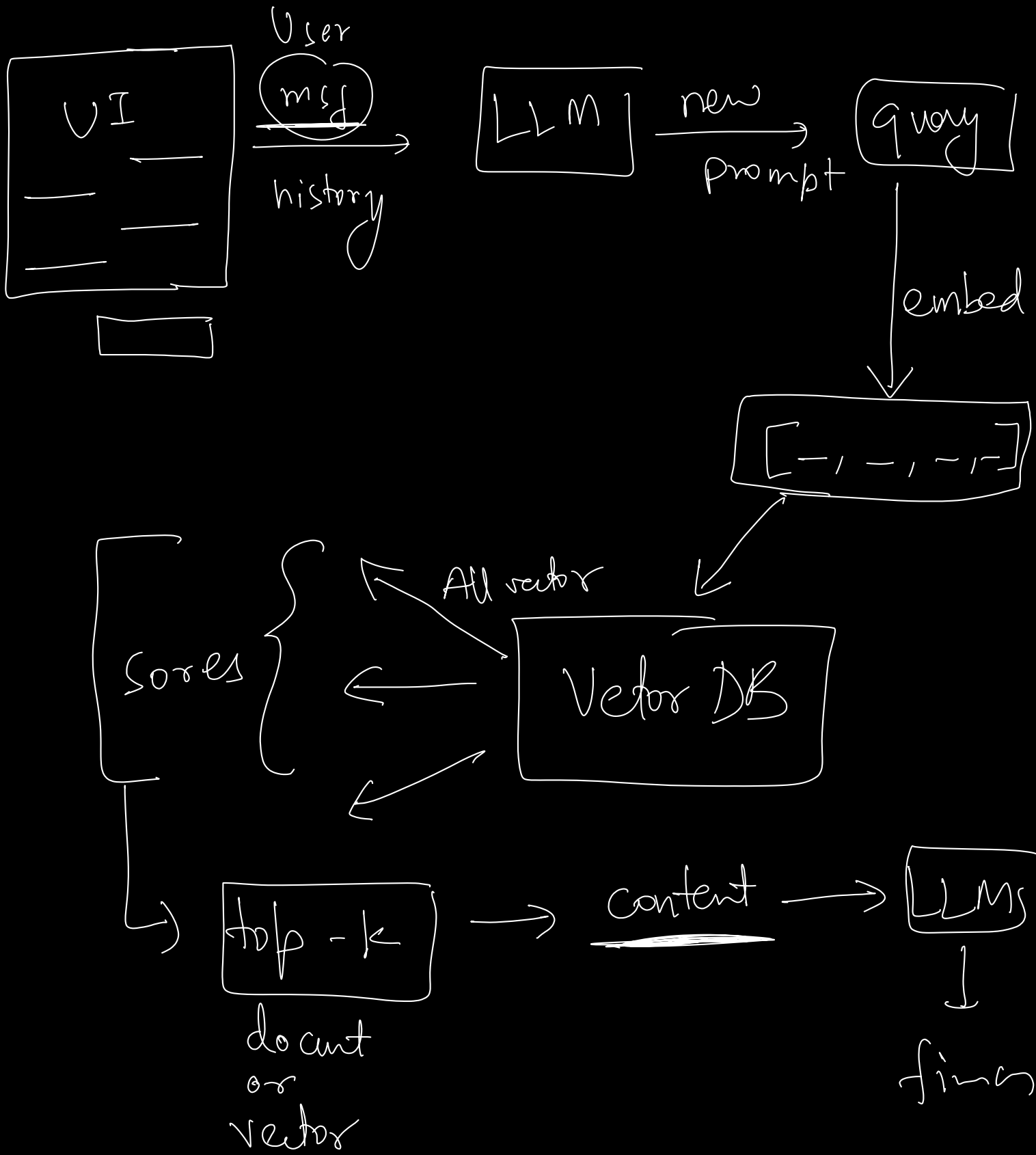


id → doc_id – (i)

vector_value → embedding [i]

metadata → doc_id
→ text_of_doc → chunk [i]
→ file_name → [ ]

⑦ RAG Retrieval



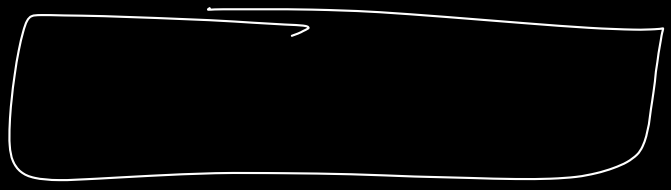
Query : Question + Chat history

Doc Ref : Via Vector DB

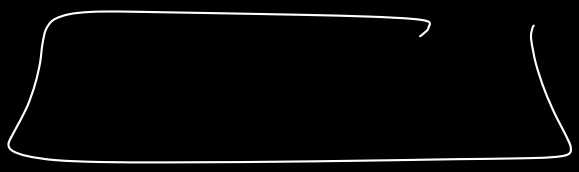
System Prm : Give answer based upon give context.

↓

LLM → answer

1. 

2. 

3. 

4. 